

Work Experience

Quantum Software Engineer

Feb 2024 - Feb 2026

QC Design

Led development of QC Design's core quantum simulation and analysis platform used directly by customers. Designed and implemented high-performance C++/Python infrastructure with custom CPython bindings, enabling **over 10,000× speedup** compared to pure Python implementations. **Prototyped and productionized advanced simulators**, error-correction codes, and optimizations for circuit analysis and logical error rate estimation. Contributed to comprehensive testing and benchmarking pipelines to ensure numerical accuracy and long-term maintainability. Worked independently while mentoring junior engineers and reviewing both code and research contributions.

Co-founder and CTO

Oct 2022 - Dec 2023

WD AI

Co-founded and served as CTO of WD AI, a **profitable AI platform** scaled to 200,000+ users and ~900 concurrent sessions. Led end-to-end development from prototype to production, building full-stack architecture, LLM pipelines, authentication, and billing. Designed and operated distributed infrastructure with CI/CD and containerization, focusing on reliability under rapid growth. Built and **led a team of 4 engineers**, overseeing hiring, architecture, and code reviews, and driving adoption of NestJS-based microservices, Kubernetes, and multi-platform integration.

Full-Stack Developer

Sep 2021 - Aug 2022

Huawei

Worked on an a **real-time collaborative 3D platform** enabling simultaneous multi-user interaction with shared virtual objects. Developed backend state synchronization and low-latency networking in C#/.NET using SignalR and REST APIs, ensuring consistent shared state across concurrent users, and implemented database-backed session and object management. Contributed to frontend interaction and scene logic in React and three.js and developed performance optimizations to improve stability and push for production, working closely with senior engineers and product management.

Developer Intern

Mar 2021 - Sep 2021

Yandex

Worked on internal backend systems for preparing and validating **large-scale datasets for LLM training and recommendation models**. Maintained and optimized C++ and Python data pipelines for daily ingestion and preprocessing of high-volume commercial data. Designed SQL-based feature extraction workflows and built automated tests that improved data quality and pipeline reliability.

Projects

Simulation of photonic quantum computing

2023

Developed a web service dedicated to simulation of linear and non-linear optics for quantum computational models using Python and Django, enabling **interactive experimentation with optical circuits**. Used [Strawberry fields](#) as an underlying engine. ([github](#))

Master's Thesis

2023-2025

Investigated a photonic-based model to implement a topological quantum memory experiment. Explored error models and **performed threshold estimations** using matching-based decoders on FBQC codes. Demonstrated how logical qubits can be reliably stored by repeatedly fusing resource states, enabling a scalable approach to fault-tolerant quantum memory.

Undergraduate Thesis

2022-2023

I did research on optimal schemes of entangling transformations in linear quantum optics. I used a genetic algorithm and ran it on **GPUs with Pytorch**. New schemes were obtained for finding the maximum entangled state, as well as for implementing gates equivalent to CX in the KLM protocol. [Presentation](#).

Education

Bachelor of Science in Computer Science and Software Engineering

Sep 2019 - Jul 2023

Saint-Petersburg State University

Master of Science in Mathematics

Sep 2022 - Nov 2025

Pontifical Catholic University of Rio de Janeiro

Technical Skills

- **Core Languages:** C++, Python, C#, TypeScript, SQL
- **Scientific & HPC:** CPython Extensions, Pybind, PyTorch, Qiskit
- **Backend & Web:** Django, React, Next.js, PostgreSQL, Redis, REST APIs
- **Infrastructure & Tools:** Git, Linux, Docker, CI/CD, Kubernetes, VPS Hosting
- **Languages:** Russian (Native), English (Fluent), Portuguese (Conversational), German (Beginner)